

Mathiyon AI — RAG Demo Document

This PDF is intentionally designed for testing Mathiyon's document upload, text extraction, chunking, retrieval, and question-answering pipeline.

1. About Mathiyon AI

Mathiyon AI is an experimental AI assistant platform. Its architecture combines a React user interface, a Node.js and Express backend, MongoDB for persistent application data, and a Python PyTorch service for language-model inference. The platform can also use a document retrieval pipeline to answer questions from uploaded files.

2. Mathematics Knowledge

For exact mathematical calculations, Mathiyon can route suitable questions to a deterministic mathematics engine. For example, $25 \times 48 = 1200$. The fraction $1/3 + 1/6 = 1/2$. The linear equation $2x + 5 = 15$ has the solution $x = 5$.

A circle with radius r has area $A = \pi r^2$. A rectangle with length l and width w has area $A = l \times w$. The derivative of x^2 is $2x$, and the integral of $2x$ with respect to x is $x^2 + C$.

3. React Development

React is a component-based JavaScript library used for building user interfaces. A React application can be divided into reusable components. State can be managed with hooks such as `useState`, while `useEffect` can be used for side effects such as data fetching.

4. RAG Test Facts

Fact	Value
Demo project name	Mathiyon AI
Primary frontend	React + TypeScript
Application backend	Node.js + Express
Database	MongoDB
AI service	Python + PyTorch
Demo model	Mathiyon SLM v1.2
Recommended retrieval Top-K	5

5. Questions to Test in Mathiyon

1. What is the name of the AI platform described in this document?
2. What database does Mathiyon use?
3. What technology is used for the AI service?
4. What is 25×48 ?
5. Solve the equation $2x + 5 = 15$.

6. What is the area formula for a circle?
7. What React hook is mentioned for managing state?
8. What is the recommended retrieval Top-K?

6. RAG Verification Challenge

If Mathiyon is asked about information that does not appear in this document, it should not invent a document-based answer. For example, ask: 'Who founded Mathiyon AI?' The correct RAG behavior is to say that the information is not present in the uploaded document unless another verified source is available.